

SOC Threat Detection & Log Analyzer

Minimum Viable Product (MVP) Document

Version 1.0

1. Introduction

The SOC Threat Detection & Log Analyzer is a cybersecurity-focused platform designed to simulate Security Operations Center (SOC) workflows. The system enables security analysts to upload and analyze log files, detect suspicious activities, classify security risks, and generate incident reports through an interactive dashboard.

The Minimum Viable Product (MVP) focuses on delivering essential threat detection and log analysis capabilities while providing meaningful security insights in an easy-to-use interface.

2. MVP Objective

The objective of the MVP is to build a functional SOC simulation platform capable of:

- Processing uploaded log files
- Detecting common attack patterns
- Identifying anomalous behavior
- Assigning risk levels
- Providing actionable security insights
- Generating downloadable incident reports

The MVP serves as a foundation for future enhancements such as machine learning-based detection, real-time monitoring, and threat intelligence integration.

3. Target Users

Security Analysts

Need:

- Log analysis tools
- Threat detection capabilities
- Incident investigation support

SOC Engineers

Need:

- Attack visibility
- Detection validation
- Security event monitoring

Cybersecurity Students

Need:

- Hands-on SOC experience
- Attack simulation environments
- Practical understanding of detection workflows

Security Researchers

Need:

- Security event analysis
- Threat intelligence insights
- Attack pattern identification

4. MVP Features

Feature 1: Log File Upload & Parsing

Description

Users can upload system log files for automated analysis.

Supported Functions

- Log ingestion
- Log parsing
- Structured event extraction

Expected Outcome

The system converts raw logs into analyzable security events.

Feature 2: Threat Detection Engine

Description

The system automatically identifies common attack patterns within uploaded logs.

Detection Capabilities

Brute Force Attack Detection

Identifies repeated authentication failures from the same source.

Suspicious Login Detection

Detects unusual login attempts and authentication anomalies.

Multi-IP Attack Detection

Identifies coordinated attacks originating from multiple IP addresses.

Time-Based Anomaly Detection

Detects activity occurring outside normal operating hours.

Geo-IP Anomaly Detection

Identifies logins originating from unexpected geographical locations.

Expected Outcome: Security alerts are generated whenever suspicious activity is detected.

Feature 3: Risk Classification Engine

Description

The system evaluates detected events and assigns risk levels.

Risk Categories

- Low Risk
- Medium Risk
- High Risk

Expected Outcome

Security teams can prioritize incidents based on severity.

Feature 4: Threat Intelligence Enrichment

Description

The platform enriches suspicious IP addresses with threat intelligence information.

Functions

- IP reputation checks
- Malicious IP identification
- Threat context generation

Expected Outcome

Analysts receive additional information about potentially malicious sources.

Feature 5: Security Dashboard

Description

A centralized dashboard provides visibility into detected threats and security events.

Dashboard Sections

Overview

Summary of security findings.

Alerts

List of detected threats.

Logs

Detailed event records.

Reports

Generated incident reports.

Expected Outcome

Users gain a comprehensive understanding of security posture and active threats.

Feature 6: Attack Simulator**Description**

Users can generate synthetic attack logs for testing detection rules.

Supported Simulations

- Brute Force Attacks
- Multi-IP Attacks
- Time-Based Anomalies
- Geo-IP Anomalies
- Combined Attack Scenarios

Expected Outcome

Users can validate detection capabilities without requiring real attack data.

Feature 7: Incident Reporting

Description

The platform generates downloadable incident reports.

Supported Formats

- TXT Reports
- PDF Reports

Expected Outcome

Security findings can be documented and shared for investigation purposes.

5. MVP Success Criteria

The MVP will be considered successful if:

- Users can upload and analyze log files.
- Threats are detected accurately.
- Risk levels are assigned appropriately.
- Security dashboards display findings correctly.
- Incident reports can be generated successfully.
- Attack simulator validates detection logic.

6. Technology Stack

Frontend

- Streamlit

Backend

- Python

Security Modules

- Custom Detection Engine
- Threat Intelligence Module
- Risk Assessment Engine

Data Processing

- Pandas
- NumPy

Visualization

- Plotly
- Streamlit Charts

Reporting

- PDF Generation Libraries

Version Control

- Git
- GitHub

Deployment

- Streamlit Cloud

7. Future Enhancements

- Real-Time Log Streaming
- SIEM Integration
- Machine Learning-Based Threat Detection
- Threat Hunting Module
- User Authentication & RBAC
- External Threat Intelligence APIs
- Cloud-Native Scalability
- MITRE ATT&CK Mapping

Conclusion

The MVP delivers the essential functionality required for security log analysis and threat detection. It enables cybersecurity professionals and learners to simulate real-world SOC operations while establishing a foundation for advanced security analytics and incident response capabilities.